The Verification Substrate Gap in Natural LLM Outputs

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Abstract

Natural language model outputs are attractive verification objects because they are portable, public, and easy to copy. They are also a weak substrate for deterministic provenance claims: the logs, proofs, model state, and decoding events that make evidence strong often do not travel with the text. We study this verification substrate gap using first-divergence token-event diagnostics. Under provider-side trace access, protected codewords are recovered in 94/96 Qwen blocks and 96/96 Llama blocks, with raw, task-only, wrong-key, and wrong-payload controls at 0/96. However, the corresponding public final text recovers zero codeword blocks. Shallow public final-text predicates show some distributional separability, but they are searchable, editable, learnable, and directly optimizable by source-mismatch examples; guided rewrite/graft attacks reach 100/100 accepted top-ranked source-mismatch rows for the tested Qwen raw, Qwen task-only, and Llama raw sources. These results support a substrate-gap claim: provider-side traces can expose controllable evidence, while current public final-text predicates do not provide public text-only verification success, natural evidence success, or an ownership proof.

1 Introduction

Commercial and open language-model ecosystems create recurring disputes about model reuse, derivative services, copied outputs, and suspected distillation. The hardest cases are not cases where no evidence can exist. Evidence can be strong when it lives in a retained protocol transcript, a cryptographic proof, a provider-side trace, model-state access, or preserved metadata. The hard case is ordinary natural final text: a user can copy it, a platform can strip metadata, and a non-cooperative service need not preserve logs or expose weights. This paper studies that gap between where strong evidence can live and what ordinary final text carries.

We call this the *verification substrate gap*. A substrate is the object that carries evidence: metadata, a protocol transcript, a proof object, provider-side event traces, model state, or the final text itself. These substrates support different verification properties. Some are deterministic but not portable; some are public but easy to spoof; some are strong only when the provider cooperates. Our question is therefore not whether evidence can be stored somewhere. It can. Our question is whether natural LLM final text can serve as a public, deterministic, portable, non-cooperative verification substrate.

The experiments reported here were originally motivated by natural-output evidence injection, but the current evidence points to a different and sharper conclusion. Provider-side first-divergence token-event traces can carry recoverable codeword evidence under reviewed controls. In the tested trace-bound diagnostics, Qwen recovers protected codewords in 94 of 96 blocks and Llama recovers them in 96 of 96 blocks, while raw, task-only, wrong-key, and wrong-payload controls remain at 0 of 96 for each model. This rules out a complete absence of model-side controllability in these diagnostics.

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The same evidence does not project into deterministic public final-text verification. Across the public final-text predicate artifacts, recovered codeword blocks remain zero. Shallow predicates can separate some protected and raw distributions, but separability is not verification. The Qwen P2 predicate has AUC 0.554676, protected row TPR 0.275476, raw row FPR 0.193455, and task-only row FPR 0.195060, with zero recovered codeword blocks. The Llama P2 predicate has AUC 0.63128, protected row TPR 0.154992, raw row FPR 0.038798, and zero recovered codeword blocks. These predicates observe weak final-text regularities, not deterministic provenance.

The final-text predicates are also attack surfaces. Once a public predicate has a nonzero accept region, source-mismatch rows can be found by rejection sampling, modified by deterministic rewrite-lite transformations, enriched by a lightweight surrogate, and directly optimized by guided rewrite/graft. The strongest guided attack reaches 100 accepted examples among the top 100 ranked source-mismatch examples for Qwen raw, Qwen task-only, and Llama raw sources. These accepts are spoofing evidence against the public final-text predicate substrate. They are not protected success, not codeword recovery, and not public text-only verification.

The paper's central thesis is therefore:

> Natural LLM outputs exhibit a verification substrate gap: provider-side first-divergence traces can carry recoverable keyed evidence, but that evidence does not become a reliable public final-text verifier. Public final-text predicates that show shallow separability fail to recover codewords and are searchable, editable, learnable, and directly optimizable by source-mismatch examples.

This thesis deliberately avoids several stronger claims. We do not claim public text-only verification success, natural evidence success, phrase-decoder success, cryptographic provenance, sanitizer robustness, payload diversity, model-family general verification, or an ownership proof. The positive evidence in this draft is a trace-bound provider-side diagnostic; the negative evidence is a public final-text observability and spoofing diagnostic.

Our contributions are:

- **A verification-substrate formulation.** We define verification axes that are often conflated: public access, deterministic recovery, portability, natural-output nativeness, non-cooperation, and spoof resistance. The current evidence supports a cooperative trace-bundle regime, not all axes simultaneously.
- **A first-divergence diagnostic.** We formalize first-divergence token events as a tokenizer-native upper-bound diagnostic for provider-side event-access settings. The diagnostic measures model-side controllability; it is not a deployable public final-text verifier.
- **An empirical substrate-gap map.** We report trace-bound recovery on Qwen and Llama with clean controls, and contrast it with zero public final-text codeword recovery.
- **A public-predicate attack ladder.** We show that shallow public final-text predicates are searchable, editable, learnable, and directly optimizable by source-mismatch examples, while explicitly not interpreting those accepted examples as protected success.
- **A claim-controlled ownership stress test.** We map seven dispute scenarios against nine method families. The current artifacts support only cooperative provider-side trace-bundle diagnostics; no current public final-text-only portable scenario succeeds.

The rest of the paper is organized as follows. Section 2 positions the work relative to watermarking, fingerprinting, provenance, and model-state methods. Section 3 defines the verifier taxonomy and threat model. Section 4 introduces first-divergence events. Sections 5–6 describe the experimental substrates and public-predicate attacks. Section 7 reports the current evidence map, and Section 8 states the substrate displacement principle and limitations.

2 Related Work

This paper is closest to work on text watermarking, LLM fingerprinting, publicly auditable watermark protocols, provenance standards, proof systems, model ownership, and extraction diagnostics. We organize the literature by the substrate that carries evidence, because substrate choice determines which verification properties can be claimed.

Text watermarking and public final text. Watermarking methods bias or test generated text so that a detector can distinguish watermarked samples from unwatermarked samples Kirchenbauer et al. [2023], Zhao et al. [2023], Hu et al. [2024], Kuditipudi et al. [2024], Christ et al. [2023], Lee et al. [2024], Chang et al. [2024], Huang and Wan [2025], Dabiriaghdam and Wang [2025], Wouters [2024], Liu and Bu [2024], Zhu et al. [2024], Lau et al. [2024], Jiang et al. [2025]. Production-scale systems such as SynthID-Text show that text watermarking can be deployed at large scale with a detector-oriented claim Dathathri et al. [2024]. This family treats final text as the main evidence object, so it is directly relevant to portability and non-cooperation. It also exposes the central substrate risk studied here: if a public final-text predicate is observable, an attacker may search, edit, paraphrase, or optimize against it. The present manuscript does not claim watermark success, public text-only verification success, or ownership proof; it studies why the tested public final-text predicates recover no codeword blocks and are spoofable.

Publicly auditable and protocol-backed watermarks. Recent work moves beyond private-key text detectors by designing publicly detectable or publicly verifiable watermark protocols Fairoze et al. [2023], Isler et al. [2023], Duan et al. [2025], Luan et al. [2026]. These systems strengthen the verification interface by adding public detectability, commitments, zero-knowledge style checks, oblivious detection, or protocol participation. In the VSG taxonomy, that is a substrate shift: the evidence is no longer ordinary final text alone, but final text plus a stronger protocol or proof object. This paper is not a replacement for those constructions. It explains why a provider-side trace-bound diagnostic should not be promoted into a final-text-only claim unless the stronger substrate is actually available.

Metadata, credentials, and proof substrates. Content provenance standards such as C2PA Content Credentials bind assertions, claims, and signatures into manifests associated with digital assets Coalition for Content Provenance and Authenticity [2025]. Proof-oriented LLM work such as zkLLM and ZKPROV studies zero-knowledge verification for model computation or dataset provenance Sun et al. [2024], Namazi et al. [2025]. These approaches make evidence stronger by moving it to a metadata, protocol, or proof substrate. We do not claim cryptographic provenance in this manuscript; we use these systems as contrast cases showing how much stronger the evidence object becomes when it does not have to survive as ordinary natural text.

Learnability, removal, and spoofing. Removal and residual-signal studies emphasize that provenance detectors should not be conflated with deterministic codeword recovery Chen et al. [2025], Sander et al. [2024]. Impossibility and attack work also shows that strong claims can fail when the attacker can transform, imitate, or learn watermark signals Zhang et al. [2023], Gu et al. [2023], An et al. [2025]. Our attack ladder has the same cautionary role, but with a narrower target: it does not claim a universal impossibility theorem over all public predicates. It shows that the tested public final-text predicates are in a weak position because they recover zero codeword blocks while still offering source-mismatch accept regions that can be searched, edited, learned, and directly optimized.

Active fingerprints and model-side behavior. Active fingerprints query a model with selected prompts and verify associated responses or behaviors Xu et al. [2024], Zeng et al. [2024a], Russinovich and Salem [2024], Pasquini et al. [2025], Yang and Wu [2024], Yamabe et al. [2025], Xu et al. [2025b,a], Bai et al. [2025], Hu et al. [2025], Nasery et al. [2025], Tsai et al. [2025]. These methods are closer to model ownership disputes than generic text watermarks because they can involve owner-chosen prompts or behaviors. Their verification substrate, however, is often a response pattern, classifier score, or query transcript rather than ordinary copied final text alone. Our work does not claim superiority over these methods. It asks a narrower diagnostic question: when evidence is visible in provider-side token events but not in public final text, what can and cannot be verified?

Model-state and training-time ownership evidence. Classical DNN watermarking embeds signatures into parameters, activations, or trigger responses Uchida et al. [2017], Zhang et al. [2018], Adi et al. [2018], Rouhani et al. [2019], Jia et al. [2021], Szyller et al. [2021], Krauß et al. [2024]. These substrates can be strong when a verifier has white-box or cooperative access, but they are less portable with copied text. Scrubbing, backdoor removal, model merging, and task-vector methods show that model-side evidence can change under post-release transformation Zhang et al. [2025], Zeng et al. [2024b], Li et al. [2024], Hubinger et al. [2024], Wortsman et al. [2022], Ilharco et al.

[2023], Yadav et al. [2023]. The present paper does not claim robustness to arbitrary fine-tuning, semantic rewriting, or probe-free distillation.

Passive provenance and extraction. Model extraction, memorization, and data-provenance studies infer exposure or lineage without necessarily injecting a recoverable owner payload Carlini et al. [2024], Liang et al. [2024], Xie et al. [2024], Puerto et al. [2025], Ravichander et al. [2025]. They motivate the non-cooperative dispute setting, but they usually answer a different question from deterministic codeword recovery. Our public final-text experiments should therefore be read as a substrate diagnostic, not as a claim that all passive provenance questions are impossible.

Positioning. The contribution here is neither a new public text-only verifier nor an ownership-proof construction. It is a claim-controlled map of which substrate currently carries evidence. Provider-side first-divergence traces show controllability under event access; public final-text predicates in the current artifacts fail to recover codewords and become source-mismatch spoofing targets. This framing complements prior methods by making explicit when a method's evidence object travels with ordinary natural text and when it requires a stronger cooperative, metadata, protocol, proof, or model-state substrate.

3 Problem Formulation and Verifier Taxonomy

We study verification as a relationship between a claim, an evidence substrate, and an access regime. The same signal can be persuasive in one regime and unusable in another. A provider-side event trace may support a cooperative audit, while copied final text may not preserve the same witness. This section defines the verifier axes used throughout the paper.

Definition 3.1 (Verification substrate)**.** A verification substrate is the object that carries or preserves evidence for a claim. Examples include platform metadata, protocol transcripts, proof objects, provider-side token-event traces, model-state access, and natural final text.

Verifier axes. We use six axes to avoid conflating different verification regimes.

Public. A third party can inspect the evidence object or run the stated test without private platform access.

Deterministic. The test recovers a codeword, proof, exact witness, or precommitted object rather than only a distributional correlation.

Portable. The evidence travels with ordinary copies or redistribution of the output.

Natural-output-native. The evidence is not an obvious Step label, fixed slot, public watermark literal, external certificate, or visible proof object.

Non-cooperative. The suspect need not provide weights, metadata, logs, special API behavior, or a trace bundle.

Hard to spoof. Public feedback does not cheaply let an attacker create accepted source-mismatch examples.

No current artifact in this paper satisfies all six axes simultaneously. The supported diagnostic result is trace-bound and cooperative: an authorized verifier has access to provider-side first-divergence token-event traces. The unsupported target is stronger: a public final-text-only method that is deterministic, portable, non-cooperative, natural-output-native, and hard to spoof. We do not claim that target is achieved.

Principle 3.1 (Substrate displacement)**.** *Moving evidence from ordinary final text to a stronger substrate can increase determinism, auditability, or spoof resistance only for access regimes in which that stronger substrate is available. A result established over a metadata, protocol, proof, trace, or model-state substrate therefore does not automatically transfer to a copied-final-text regime.*

Principle 3.1 is a scope rule rather than an impossibility theorem. It is the reason we separate trace-bound diagnostics from public final-text predicates throughout the manuscript. Trace-bound codeword recovery can demonstrate provider-side event controllability, but it does not become a public final-text claim when the trace is absent.

Substrate	Public	Deterministic	Portable	Non-coop.	Current role
Metadata	Sometimes	Sometimes	Fragile	Sometimes	Strong only if preserved
Protocol transcript	Usually	Often	No	No	Strong under participation
Proof object	Yes	Yes	Conditional	Conditional	Requires a statement and proof
Provider trace	No	Yes	No	No	Supported cooperative diagnostic
Model state	No	Often	No	No	White-box or cooperative evidence
Natural final text	Yes	Not shown	Yes	Yes	Current public substrate fails codeword recovery

Table 1: Verification substrates differ in which properties they can support. The table is a taxonomy, not a ranking. Current results support the provider trace row under cooperative event access; they are not public text-only verification success.

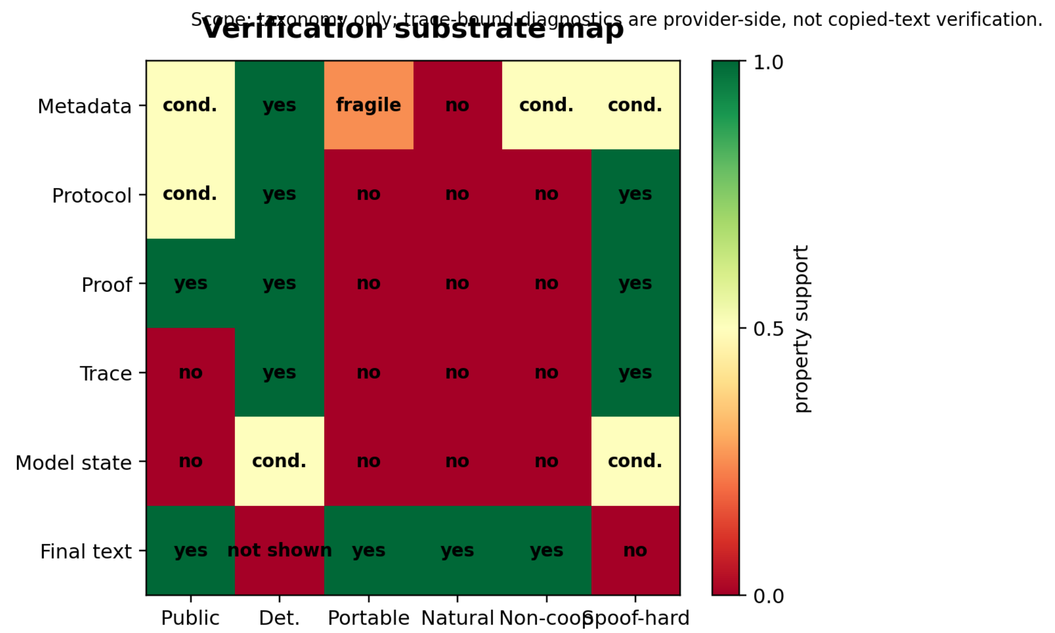

Figure 1: Verification substrate map. The map summarizes which properties different substrates can support. It is a taxonomy and does not claim a universal impossibility theorem; it is not public text-only verification success and not an ownership proof.

Threat model. The public final-text setting assumes an observer sees only the generated text. The observer does not receive the model weights, hidden states, logits, provider logs, trace bundles, or cryptographic signatures. The attacker may query a public predicate, search over source-mismatch examples, apply simple rewrites, learn a surrogate ranker from public feedback, and optimize text against the predicate. The attacker is not assumed to know a private provider key for the trace-bound diagnostic.

Source-mismatch examples. A source-mismatch example is a row produced by a source that should not satisfy the protected claim, such as raw, task-only, wrong-key, or wrong-payload text. If a public predicate accepts such a row, that acceptance is spoofing evidence against the predicate. It is not protected success, not codeword recovery, not natural evidence success, and not an ownership proof.

Current claim boundary. The supported claim is limited to the following statement: reviewed Qwen and Llama trace-bound diagnostics show provider-side event controllability under authorized trace access, while public final-text predicates in the current artifacts recover zero codeword blocks and are vulnerable to source-mismatch optimization. The paper does not claim public text-only verification success, cryptographic provenance, signed trace provenance, sanitizer robustness, payload-diversity evidence, or model-family general verification.

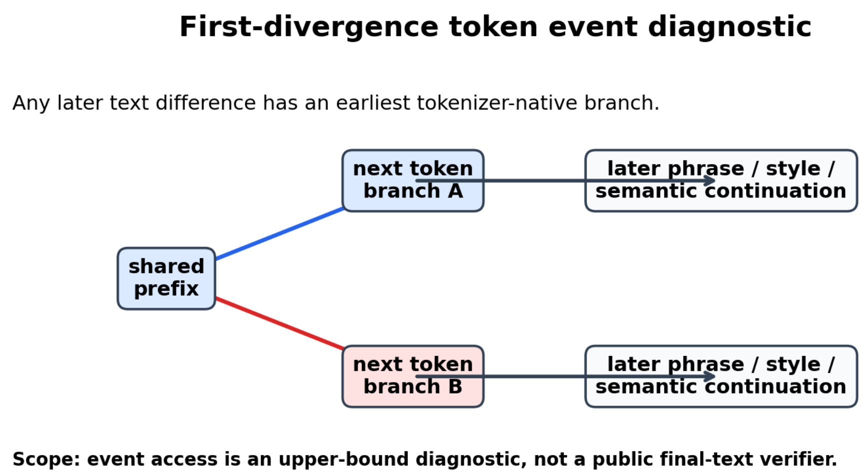

Figure 2: First-divergence token events as a diagnostic unit. The figure shows how two continuations share a prefix before branching at a next-token event. This is a trace-bound controllability diagnostic, not a public final-text verifier and not public text-only verification success.

4 First-Divergence Token Event Diagnostic

The trace-bound diagnostic is based on a simple property of autoregressive token sequences. If two continuations differ as text, then after their longest shared token prefix there is a first next-token choice at which they diverge. We use that event as a diagnostic unit because it is the earliest tokenizer native point at which a phrase, style, template, or semantic trajectory can become distinguishable.

Definition 4.1 (First-divergence event)**.** Let $u = (u_1, \ldots, u_m)$ and $v = (v_1, \ldots, v_n)$ be two token continuations after a shared prompt prefix. If $u \neq v$, their first-divergence position is the smallest index t for which $u_t \neq v_t$, treating end-of-sequence as a token when one sequence is a strict prefix of the other. The first-divergence event is the next-token branch observed at that position.

Lemma 4.1 (First divergence of text differences)**.** *For any two distinct tokenized continuations produced by an autoregressive tokenizer-based model, there exists a first-divergence position after their longest shared token prefix.*

Proposition 4.2 (First-divergence reduction)**.** *Let a deterministic evidence rule partition tokenized continuations into two or more observable classes. If two continuations fall in different classes, then the class distinction has at least one first-divergence event witness: after the continuations' longest shared token prefix, their next-token branches are different.*

Proposition 4.3 (Public-predicate spoofability under nonzero source density)**.** *Let $A(x) \in \{0, 1\}$ be a deterministic public predicate over final text, and let D be a source-mismatch proposal distribution. If $p = \Pr_{x \sim D}[A(x) = 1] > 0$, then rejection sampling from D finds an accepted source-mismatch example with expected $1/p$ predicate evaluations. Such an accepted example is spoofing evidence against the predicate, not protected success or codeword recovery.*

Lemma 4.1 is elementary, but it clarifies the role of the diagnostic. Proposition 4.2 extends the same observation from pairwise text differences to deterministic evidence classes: any class distinction that appears in tokenized continuations must be witnessed by some earliest branch. If a provider-side trace records those events and binds them to the final output, then the trace can expose information that may be absent from a public final-text view.

Upper-bound role. The first-divergence diagnostic is an upper-bound measurement of model-side controllability under event access. If protected codewords cannot be recovered from first-divergence traces, then a weaker public final-text-only substrate is unlikely to recover the same keyed evidence. If first-divergence traces do recover codewords while public final text recovers none, the bottleneck is public observability rather than the absence of event-level capacity.

Layer	Evidence object	Supported wording	Forbidden wording
Trace-bound diagnostic	Provider-side first-divergence events	Event-access controllability under reviewed controls	Public text-only verification success
Public final-text predicate	Final generated text only	Shallow observability baseline; zero codeword recovery	Phrase-decoder success or natural evidence success
Attack ladder	Source-mismatch rows plus public predicate feedback	Predicate spoofing evidence	Protected success or codeword recovery
Ownership stress test	Scenario-method matrix	Cooperative trace-bundle support only	General ownership proof

Table 2: Evidence map by substrate. Each layer has a different claim scope. The analysis keeps source-mismatch public-predicate accepts separate from protected trace-bound recovery.

Not a public verifier. The diagnostic is not a deployable public final-text verifier. It assumes provider-side event access and therefore belongs to the trace substrate in Table 1. It can support a cooperative trace-bundle audit, but it does not by itself support copied-text verification, rewritten-text verification, cryptographic provenance, or an ownership proof.

Why this matters for the experiments. The experiments below compare two views of the same broad phenomenon. In the trace-bound view, an authorized verifier observes selected first-divergence token events and tests whether a protected codeword is recoverable under controls. In the public final-text view, a third party observes only text and tries to recover or approximate the signal from shallow predicates. The verification substrate gap is the empirical separation between these views.

Why shallow predicates are not enough. Proposition 4.3 gives the minimal reason a public final-text predicate with nonzero source-mismatch density is risky. It does not say that every public predicate is breakable, and it is not a universal impossibility theorem. It says that a predicate with an observable accept region becomes a target for search. In the current artifacts, the Qwen raw P2 source-mismatch rate is 0.193455, the Qwen task-only rate is 0.195060, and the Llama raw rate is 0.038798. A direct rejection sampler would therefore expect accepted source-mismatch examples after about 5.17, 5.13, and 25.8 predicate evaluations respectively, before any rewrite, surrogate ranking, or guided optimization is applied.

5 Experimental Substrates and Evidence Map

This section describes the three experimental substrates used in the current paper architecture. The purpose is to keep the claim tied to the substrate that actually carries the evidence.

Trace-bound diagnostic substrate. The trace-bound corpus records provider-side first-divergence events under protected and control arms. The protected arm tests whether the reviewed event route can carry the expected codeword. The raw, task-only, wrong-key, and wrong-payload arms test whether the same recovery appears without the protected route or under mismatched credentials. The claim scope is provider-side event controllability only.

Public final-text predicate substrate. The public final-text corpus removes trace access and evaluates predicates over final text. The strongest current public predicates show shallow distributional separability, but they recover zero codeword blocks. Their role is diagnostic: they test whether final text exposes enough signal for deterministic codeword recovery. In the current artifacts, it does not.

Public-predicate attack substrate. The attack ladder treats a public final-text predicate as a feedback function. It tests whether source-mismatch examples can be found, edited, learned, or optimized into the predicate's accept region. Success here is not protected success. It is evidence that the predicate is spoofable and therefore cannot be treated as deterministic provenance.

Quality gates. Earlier routes failed when visible structures, duplicate outputs, or forbidden public literals made the result unsuitable as natural-output evidence. Those failures remain part of the historical evidence chain. The VSG framing does not reclassify them as successes. Instead, it uses them to motivate why the final text substrate must be audited separately from trace-bound controllability.

Claim family	Status	Evidence	Boundary
Trace-bound first-divergence recovery	Supported diagnostic only	Qwen 94/96, Llama 96/96, controls 0/96	Not public text-only verification or ownership proof
Public final-text codeword recovery	Not supported	Recovered codeword blocks = 0	No phrase-decoder success
Public predicates as attack targets	Supported spoofing evidence only	Source-mismatch attacks succeed across the ladder	Not protected success
Copied or rewritten final-text disputes	Not supported	Stress test has no public final-text-only portable success scenario	No non-cooperative copied-text resolution

Table 3: Claim ledger for the current evidence map. Each row pairs allowed evidence with a boundary so the draft does not drift into public text-only verification success or ownership proof.

Ownership scenario stress test. The current stress test contains 63 cells, from seven dispute scenarios crossed with nine method families. Only the cooperative provider with trace-bundle scenario is supported by current artifacts. No current public final-text-only portable scenario succeeds. This matrix prevents the trace-bound diagnostic from being overstated as a general non-cooperative ownership solution.

6 Public Final-Text Predicates and Attack Model

The public final-text experiments ask whether a third party, given only text, can recover a deterministic codeword or at least build a reliable public predicate for provenance. The answer in the current artifacts is negative: the predicates recover zero codeword blocks. The same predicates nevertheless have nonzero accept regions, which makes them targets for source-mismatch optimization.

Public predicate. A public final-text predicate is any fixed rule in the current artifact set that maps final text to a score or accept/reject decision without using provider-side traces. The P2 predicates used in the results are shallow surrogates for final-text observability. They are not claimed to exhaust all possible public predicates, and they are not claimed to be deployment-grade verifiers. Their diagnostic role is narrower: if they recover no codeword blocks, they do not verify; if they still accept source-mismatch rows, their accept region is spoofable.

Attack ladder. We evaluate four increasingly strong public-predicate attacks:

Rejection sampling. Query the public predicate on existing source-mismatch rows and select accepts.

Rewrite-lite. Apply deterministic opener or phrasing edits and requery the predicate.

Distillation-lite. Train or fit a lightweight surrogate ranking signal from public predicate feedback, then rank source-mismatch rows.

Guided rewrite/graft. Use public predicate feedback to choose and graft predicate-favorable text fragments onto source-mismatch rows.

These attacks do not prove that every public final-text predicate is broken. They show that the strongest current final-text surrogate predicates occupy a bad regime: they do not recover codewords, but their nonzero accept regions can be found or optimized by source-mismatch examples.

Selection-bias boundary. Attack success must not be confused with protected recovery. The attack pipeline selects or modifies source-mismatch text using public predicate feedback. It does not use the protected trace route, does not recover the protected codeword, and does not establish natural-output provenance. Its purpose is to stress the public final-text predicate substrate.

Naturalness boundary. The current guided rewrite/graft attacks are artifact-level predicate attacks. Their accepted source-mismatch examples should not be described as human-level natural or semantically equivalent unless a separate naturalness and semantic preservation evaluation is run. This limitation weakens any claim about attack realism, but it does not rescue a predicate that fails codeword recovery and accepts source-mismatch rows.

Model	Protected	Raw	Task-only	Wrong-key	Wrong-payload
Qwen	94/96	0/96	0/96	0/96	0/96
Llama	96/96	0/96	0/96	0/96	0/96

Table 4: Trace-bound first-divergence diagnostic results. These numbers support provider-side event controllability under trace access. They do not support public final-text-only verification success or an ownership proof.

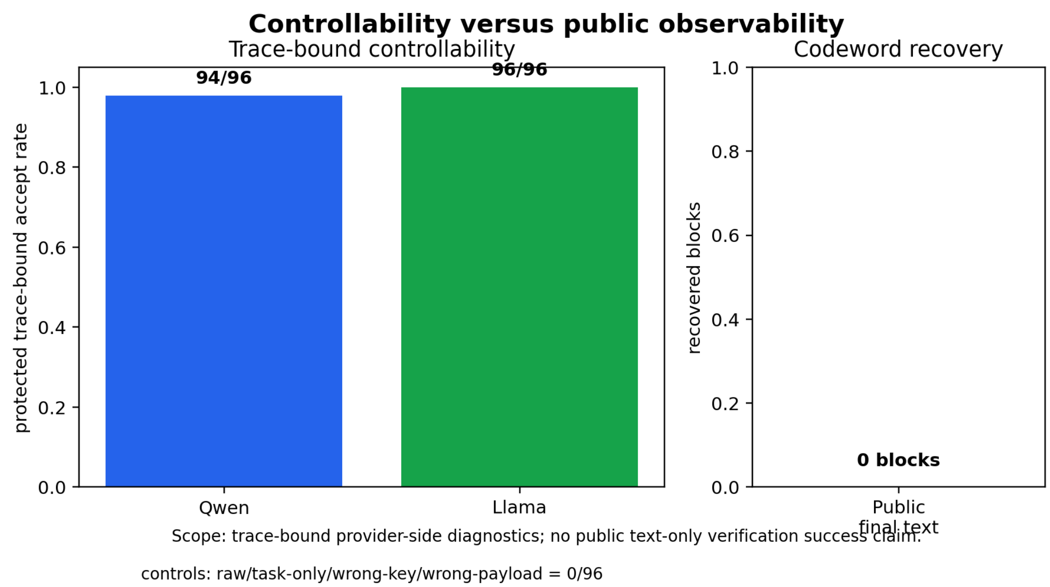

Figure 3: Controllability versus public observability. Trace-bound protected accepts are high with clean controls, while public final-text codeword recovery is not observed. The figure supports a substrate-gap diagnosis, not public text-only verification success.

7 Evaluation

The evaluation is organized by substrate rather than by historical run name. It reports what the current artifacts support and what they do not support. All public final-text source-mismatch accepts below are spoofing evidence only; they are not protected success, not codeword recovery, and not public text-only verification.

7.1 Trace-Bound Controllability

The trace-bound diagnostic tests provider-side first-divergence event access. Table 4 reports protected recovery and the four control arms.

The protected recovery rates show that the event-access substrate can carry a recoverable codeword under reviewed controls. The clean control arms rule out the simplest raw, task-only, wrong-key, and wrong-payload explanations in this diagnostic. The result is deliberately scoped to trace-bound provider-side access.

7.2 Public Final-Text Observability

The public final-text route removes event access. Across the current public predicate artifacts, the number of recovered codeword blocks is zero. The P2 predicates show shallow separability, summarized in Table 5, but no deterministic codeword recovery.

This result is the core observability gap. The trace-bound signal does not become a public final-text codeword. A public predicate can separate distributions without verifying provenance.

Model	AUC	Protected TPR	Raw FPR	Task-only FPR	Codeword blocks
Qwen	0.554676	0.275476	0.193455	0.195060	0
Llama	0.631280	0.154992	0.038798	–	0

Table 5: Public final-text P2 predicate diagnostics. The predicates show some protected/raw separability, but codeword recovery remains zero. AUC is not deterministic verification.

Source	Arm	Rejection top-100	Distillation top-100	Guided top-100
Qwen	raw	14/100	51/100	100/100
Qwen	task-only	20/100	38/100	100/100
Llama	raw	5/100	20/100	100/100

Table 6: Public-predicate attack ladder on source-mismatch rows. Guided rewrite/graft reaches 100/100 accepted top-ranked examples for all three listed source arms, at 3,400 predicate queries per top-100 source set. These are source-mismatch spoofing examples, not protected success.

7.3 Public-Predicate Spoofing

The attack ladder evaluates whether source-mismatch rows can enter the public predicate's accept region. Table 6 summarizes rejection sampling, distillation-lite ranking, and guided rewrite/graft.

Rewrite-lite also produces thousands of new source-mismatch accepts under deterministic opener edits: 6,232 prepend-opener and 7,108 replace-opener new accepts for Qwen raw; 6,214 and 6,957 for Qwen task-only; and 7,973 and 9,384 for Llama raw. Distillation-lite enriches top-ranked source-mismatch accepts: Qwen raw reaches 51/100 with 2.318x enrichment, Qwen task-only reaches 38/100 with 2.533x enrichment, and Llama raw reaches 20/100 with 20.0x enrichment.

The interpretation is narrow but important. These attacks do not show protected recovery. They show that public final-text predicates with nonzero accept regions are optimization targets.

7.4 Template Leakage and Duplicate Quality

Earlier natural-output routes exposed quality problems that matter for the substrate claim. The template leakage audit reports Qwen shallow AUC 0.554676, which passes the 0.60 gate, but a format-scrub duplicate extra row remains. Llama reports shallow AUC 0.63128, which fails the 0.60 leakage gate. These quality failures are not repaired into positives. They are evidence that final text can carry visible or brittle regularities without becoming a reliable verification substrate.

7.5 Ownership Scenario Stress Test

The scenario stress test crosses seven ownership scenarios with nine method families, for 63 cells. The only currently supported scenario is the cooperative provider with trace bundle. The public final-text-only portable supported set is empty. This result is not a survey claim about all future methods; it is a claim-boundary check for the current artifacts.

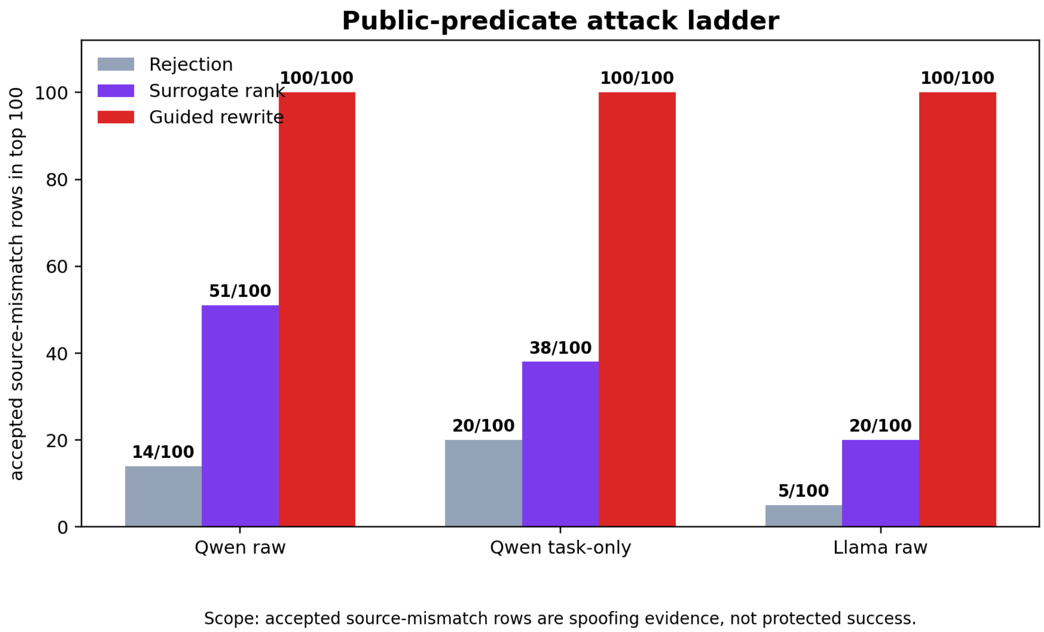

Figure 4: Public-predicate attack ladder. Accepted rows in this figure are source-mismatch spoofing examples produced under public predicate feedback. They are not protected success, not codeword recovery, and not public text-only verification.

Stress-test quantity	Current value
Scenario-method cells	63
Supported trace-bound scenarios	1
Supported public final-text-only portable scenarios	0
Supported scenario label	Cooperative provider with trace bundle

Table 7: Ownership scenario stress-test summary. Current artifacts support a cooperative trace-bundle diagnostic only.

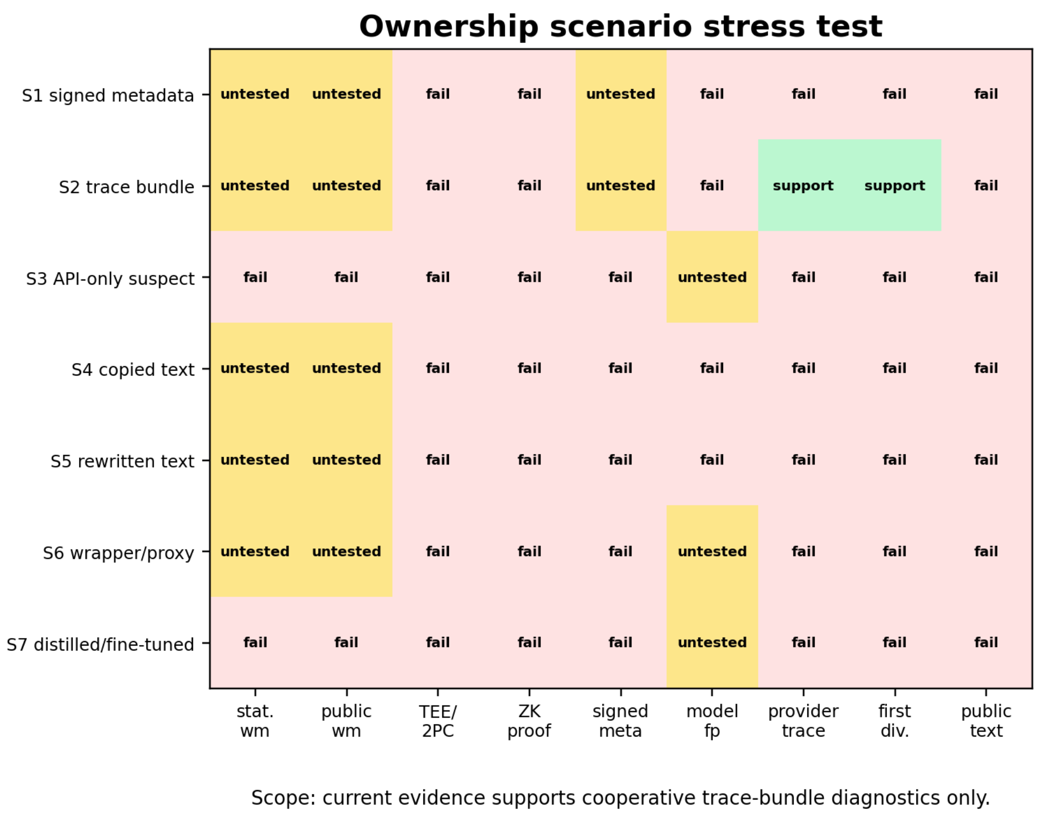

Figure 5: Ownership scenario heatmap. The current artifacts support only the cooperative provider trace-bundle diagnostic scenario; no public final-text-only portable success scenario is supported, and no ownership proof is claimed. This is not public text-only verification success and not an ownership proof.

8 Discussion and Limitations

Substrate displacement. The central pattern in the evidence is substrate displacement. Verification becomes stronger when evidence is moved to a stronger substrate, but stronger substrates often travel less naturally with ordinary LLM outputs. Metadata can be strong if preserved, but it is fragile under copy-paste or platform stripping. Protocol transcripts and proof objects can be deterministic, but they require participation and a committed statement. Provider-side traces can recover codewords under authorized event access, but they do not travel with public final text. Natural final text is portable and non-cooperative, but the current public predicates either fail codeword recovery or become source-mismatch spoofing targets.

No universal impossibility theorem. The paper does not prove that every conceivable public final-text method must fail. It reports a conditional empirical and diagnostic result: in the tested artifacts, provider-side first-divergence traces carry recoverable evidence, while public final-text predicates recover zero codeword blocks and expose spoofable accept regions. Stronger public final-text schemes would need to show deterministic recovery and spoof resistance under their own declared substrate and threat model.

P2 and P3 are surrogates, not exhaustive classes. The shallow predicates used in the public final-text experiments are surrogate attempts to recover or approximate protected signal from final text. They are useful because they occupy the diagnostic failure mode of interest: no codeword recovery, but a nonzero accept region. We do not claim that these predicates exhaust all possible public final-text verifiers.

Trace-bound success is not deployment proof. The trace-bound diagnostic measures event-access controllability. It should not be described as public text-only verification success, natural evidence success, phrase-decoder success, signed provenance, cryptographic provenance, or an ownership proof. A deployable cooperative trace-bundle system would also need trace binding, key management, audit policy, and release engineering that are outside the present claim.

Attack naturalness is not yet established. The guided rewrite/graft attacks show public-predicate optimizability. The current artifact-level result does not claim that every rewritten example is semantically preserved, human-natural, or indistinguishable from ordinary answers. Adding a naturalness or semantic-preservation filter could strengthen the attack, but it is not required to show that a predicate with zero codeword recovery and source-mismatch accepts is not a deterministic verifier.

Historical failures remain failures. Earlier routes failed for important reasons: passive evidence lacked enough observable symbols, step-labeled outputs exposed brittle structure, phrase decoders failed under scrubbed final text, and natural-output diagnostics surfaced duplicate or forbidden-quality failures. The VSG framing does not reclassify those artifacts as positives. It explains why the paper should separate trace-bound controllability from public final-text verification.

Unclaimed axes. The current artifacts do not claim sanitizer robustness, robustness to semantic paraphrase, full false-accept-rate guarantees, payload-diversity evidence, Llama public-text transfer, model-family general verification, copied-text ownership resolution, rewritten-text ownership resolution, or non-cooperative ownership proof. These are future work or out-of-scope axes, not hidden successes.

9 Conclusion

This paper reframes the current evidence around a verification substrate gap. Provider-side first-divergence traces can carry recoverable keyed evidence under reviewed event-access conditions: the tested Qwen and Llama diagnostics recover protected codewords while rejecting controls. The same evidence does not become a deterministic public final-text codeword in the current artifacts. Public final-text predicates show shallow separability, but they recover zero codeword blocks and can be searched, edited, learned, and optimized by source-mismatch examples.

Route	Status	Bottleneck	Use in VSG
v1 passive mining	Historical failure	Public-symbol survival and observability failed	Motivates substrate-gap framing only
R3 Step-label/fixed-slot route	Historical failure	Visible structure and slot fragility	Not reclassified as positive
R4 phrase/final-text route	Historical failure	Public final-text codeword recovery remains zero	Supports observability gap
R4 trace-bound route	Diagnostic success under trace access	Not public final-text verification	Upper-bound controllability evidence only

Table 8: Historical failure chain motivating the substrate-gap framing. The chain does not imply public final-text verification success or ownership proof.

The resulting claim is intentionally narrower than an ownership-verification claim. The work supports cooperative trace-bound diagnostics and a negative public final-text substrate diagnosis. It does not support public text-only verification success, natural evidence success, phrase-decoder success, cryptographic provenance, or an ownership proof. The main lesson is that evidence strength depends on where the evidence lives. Stronger substrates can make verification more deterministic, but they may not travel with ordinary natural LLM outputs; final text travels easily, but current public predicates do not provide reliable deterministic verification.

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A Proof Sketches

A.1 Proof of Lemma 4.1

Proof. Let $u = (u_1, \ldots, u_m)$ and $v = (v_1, \ldots, v_n)$ be two distinct token continuations. If neither sequence is a prefix of the other, then the set

$$\{t : 1 \leq t \leq \min(m, n),\ u_t \neq v_t\}$$

is nonempty, and the least element is the first-divergence position. If one sequence is a strict prefix of the other, then the first position after the shorter sequence ends is the first divergence when end-of-sequence is treated as a token. Thus every pair of distinct tokenized continuations has a first divergence after their longest shared token prefix. □

A.2 Diagnostic Consequence

The lemma does not state that final text exposes a reliable public verifier. It only states that any token-level difference has an earliest divergent event. This is why first-divergence traces are useful as an upper-bound diagnostic: if a provider-side event trace cannot recover a signal, then the weaker public final-text view is unlikely to recover that same signal. If the trace recovers a signal while public final text does not, the observed bottleneck is public observability rather than event-level controllability.

B Formal Substrate-Gap Propositions

This appendix expands the formal statements in Section 4. The statements are intentionally narrow. They support the trace-bound diagnostic framing and the spoofability interpretation of the tested public predicates; they do not prove a universal impossibility theorem for all future public final-text predicates.

B.1 First-Divergence Reduction

Proof of Proposition 4.2. Let C be the deterministic evidence rule, and let u and v be tokenized continuations such that $C(u) \neq C(v)$. Since a deterministic rule assigns the same class to identical inputs, u and v cannot be identical token sequences. By Lemma 4.1, distinct token sequences have a first-divergence position after their longest shared prefix. The two branches at that position are therefore a witness that the continuations have already become distinguishable at tokenizer-native next-token resolution. □

The proposition does not say that the first divergent token alone is sufficient to decode a payload or verify a claim. It only states that any deterministic class distinction between two tokenized continuations must begin at some next-token branch. This is why first-divergence traces are useful as an upper-bound diagnostic for provider-side event controllability.

B.2 Source-Mismatch Accept Density

Proof of Proposition 4.3. Draw independent samples $x_1, x_2, \ldots$ from the source-mismatch proposal distribution D and evaluate the deterministic public predicate A on each sample. Each trial succeeds with probability $p = \Pr_{x \sim D}[A(x) = 1]$. The number of trials until the first accepted sample is therefore a geometric random variable with expectation $1/p$ when $p > 0$. The accepted sample came from D, a source that should not satisfy the protected claim. Its acceptance is therefore evidence that the predicate has a source-mismatch accept region. It is not protected success, not codeword recovery, and not a trace-bound recovery result. □

The result is deliberately conditional on the tested predicate and proposal distribution. It does not rule out every possible future public predicate. It explains why the predicates evaluated in this manuscript must be treated as spoofing surfaces when they have nonzero source-mismatch accept density.

B.3 Substrate Displacement Scope

Principle 3.1 is a claim-scope rule. A metadata manifest, protocol transcript, proof object, provider trace, or model-state witness can carry stronger evidence than ordinary final text. However, the stronger result applies only when the corresponding substrate is available to the verifier. A copied-text dispute that lacks that substrate cannot inherit the same deterministic witness merely because the stronger substrate exists elsewhere.

This principle is not a no-go theorem for all verification systems. It is a discipline for reading the experiments in this paper: trace-bound recovery is evidence of provider-side event controllability, while public final-text predicate accepts from source-mismatch examples are evidence of spoofability in the tested public-predicate substrate.

C Public-Predicate Attack Example Audit

This appendix records representative source-mismatch rows from the public-predicate-guided rewrite/graft artifact. The examples are included to show how a public final-text predicate can be optimized when it has a nonzero accept region. They are not protected success, not codeword recovery, and not a naturalness claim.

The corresponding review artifact records three representative rows: Qwen raw, Qwen task-only, and Llama raw. Each row is selected from rank 1 guided rewrite/graft examples and records the original score, rewritten score, threshold, query cost, source arm, and a report-only naturalness caveat. The artifact does not run a model or a new evaluator. It only summarizes existing source-mismatch rows from the attack ladder.

D Extended Related Work and Substrate Positioning

This appendix expands Section 2. It is a substrate-positioning map, not a second contribution section and not an ownership-proof claim.

Model / arm	Attack	Score change	Threshold	Original excerpt	Rewritten excerpt
Qwen raw	Guided rewrite/graft	15.603 → 43.926	2.907	Helpful start: read and understand communication guidelines.	Low-risk start: organize a meeting where everyone introduces skills, then graft the original communication-guideline content.
Qwen task-only	Guided rewrite/graft	19.051 → 46.825	2.907	Helpful start: standardize templates for common responses.	Low-risk start: organize a meeting where everyone introduces skills, then graft the original template-standardization content.
Llama raw	Guided rewrite/graft	32.384 → 64.473	9.269	Small practical step: schedule a household meeting to discuss workshop feedback.	Helpful start: organize a committee into smaller groups, then graft the original household-meeting content.

Table 9: Representative public-predicate-guided source-mismatch examples. The rewrites cross the public predicate threshold, but no semantic naturalness gate was applied and the accepted rows remain spoofing evidence only.

Generated-text watermarking. Token-level watermarking methods mark generated text so that a detector can identify samples produced under a watermarking rule Kirchenbauer et al. [2023], Zhao et al. [2023], Hu et al. [2024], Kuditipudi et al. [2024], Christ et al. [2023], Lee et al. [2024], Chang et al. [2024], Huang and Wan [2025], Dabiriaghdam and Wang [2025], Wouters [2024], Liu and Bu [2024], Zhu et al. [2024], Lau et al. [2024], Jiang et al. [2025]. SynthID-Text is an important production-scale example because it reports a large deployment and detector evaluation over generated text Dathathri et al. [2024]. These methods use final text as the evidence object, making them highly relevant to public and portable detection. The VSG result does not claim watermark failure in general. It shows that the current public final-text predicates in our artifacts recover zero codeword blocks and are vulnerable to source-mismatch spoofing.

Publicly auditable watermark protocols. Publicly-Detectable Watermarking, Puppy, PVMark, and VOW strengthen the verification interface by adding public detectability, public verifiability, commitments, oblivious detection, or zero-knowledge style protocol structure Fairoze et al. [2023], Isler et al. [2023], Duan et al. [2025], Luan et al. [2026]. These works are central to the VSG comparison because they show that public audit can be pursued by changing the evidence substrate. Our manuscript should not be read as a claim against those protocol-backed routes. It asks what remains when the verifier sees only ordinary final text and no protocol or proof object.

Credential and proof systems. C2PA Content Credentials bind assertions, claims, and signatures in a manifest associated with a digital asset Coalition for Content Provenance and Authenticity [2025]. zkLLM and ZKPROV explore proof-based verification for LLM computation or dataset provenance Sun et al. [2024], Namazi et al. [2025]. These systems are strong examples of substrate displacement: they can support deterministic checks when the credential, statement, proof, or transcript is available. We do not claim cryptographic provenance in this manuscript; we use these systems to mark the difference between proof-bearing substrates and ordinary copied natural text.

Removal, rewriting, and learnability. Watermark removal and residual-signal work shows that text-level provenance can be changed, removed, inherited, or transformed under downstream processing Chen et al. [2025], Sander et al. [2024]. Impossibility, learnability, and spoofing studies further caution that watermark signals can be unstable or learnable under some attacker models Zhang et al. [2023], Gu et al. [2023], An et al. [2025]. This literature motivates our separation between shallow distributional separability and deterministic codeword recovery. The current manuscript does not claim sanitizer robustness or semantic-paraphrase robustness; those would require separate gates.

Active fingerprints. LLM fingerprinting methods use prompts, response patterns, hidden triggers, human-readable signatures, chained responses, cross-turn correlations, model identification behavior, probabilistic traces, and scalable verification Xu et al. [2024], Zeng et al. [2024a], Russinovich and Salem [2024], Pasquini et al. [2025], Yang and Wu [2024], Iourovitski et al. [2024], Yamabe et al. [2025], Xu et al. [2025b,a], Bai et al. [2025], Hu et al. [2025], Nasery et al. [2025], Tsai et al. [2025]. These systems are closest to model-side verification. Our paper should not be read as a superiority

Method family	Evidence substrate	What it can support	Needed assumption	VSG positioning
Private-key text watermarks	Final generated text plus detector key	Distributional detection or keyed detection	Text remains sufficiently unmodified; detector has key	Closest final-text substrate; current artifacts do not claim watermark success
Publicly auditable watermarks	Final text plus public protocol, commitment, or proof object	Publicly auditable detection under protocol assumptions	Protocol artifacts or public verification material are available	Stronger substrate than ordinary final text alone
C2PA-style credentials	Asset metadata manifest plus signatures	Manifest integrity and asserted provenance metadata	Credential survives copying and trust anchors are accepted	Metadata substrate; not the copied-text-only regime
ZK/proof systems	Statement, witness, and proof transcript	Deterministic proof of specified computation or provenance statement	Statement is well-formed and proof substrate is available	Proof substrate; not claimed by current trace-bound artifacts
Provider traces	Token-event logs or first-divergence traces	Cooperative provider-side controllability diagnostics	Authorized verifier receives trace bundle	Supported only as trace-bound diagnostic in current results
Model-state fingerprints	Weights, activations, or trigger behavior	White-box or cooperative ownership evidence	Model access or cooperative query conditions	Complementary model-state substrate
Public final text only	Ordinary copied natural text	Portable observation; in current artifacts zero codeword recovery	No logs, proofs, metadata, or model access	Current tested predicates are spoofable and not deterministic verification

Table 10: Related work positioned by evidence substrate. The table is a claim scope map, not a ranking and not an ownership-proof claim. The VSG result is that trace-bound evidence can be recoverable while the tested public final-text predicates recover no codeword blocks and can be spoofed by source-mismatch examples.

comparison. It separates the substrate question: does the evidence live in a cooperative query/trace regime, or does it survive as a deterministic witness in public final text?

Model-state ownership evidence. Classical DNN watermarking and ownership methods embed signatures in parameters, activations, or trigger responses Uchida et al. [2017], Zhang et al. [2018], Adi et al. [2018], Rouhani et al. [2019], Jia et al. [2021], Szyller et al. [2021], Krauß et al. [2024]. These substrates can support strong claims under white-box or cooperative access. They do not automatically solve the public final-text-only setting in which a third party has only copied natural text.

Post-release modification. Fingerprint erasure, backdoor removal, sleeper-agent persistence, LoRA, weight averaging, task arithmetic, and model merging all show that downstream model behavior can change after release Zhang et al. [2025], Zeng et al. [2024b], Li et al. [2024], Hubinger et al. [2024], Hu et al. [2021], Wortsman et al. [2022], Ilharco et al. [2023], Yadav et al. [2023]. This motivates scenario-specific evaluation rather than broad robustness language. The current VSG manuscript does not claim robustness to arbitrary fine-tuning, merging, semantic rewriting, or probe-free distillation.

Passive provenance and exposure. Passive provenance methods infer exposure, memorization, or lineage without an owner-controlled active evidence channel Carlini et al. [2024], Liang et al. [2024], Xie et al. [2024], Puerto et al. [2025], Ravichander et al. [2025]. They are complementary to the VSG framing. They can support forensic analysis in some regimes, but they do not by themselves establish that ordinary final text carries a deterministic codeword witness.

Substrate takeaway. Across these families, strong evidence usually depends on the substrate being available. The present manuscript's contribution is to make that dependence explicit: trace-bound first-divergence diagnostics show provider-side event capacity, while current public final-text predicates do not recover codewords and can be optimized by source-mismatch examples.

E Reproducibility and Artifact Scope

This appendix records the artifact scope for the VSG manuscript snapshot. It does not disclose private cluster paths, credentials, or secret keys. It also does not convert trace-bound diagnostics into public text-only verification claims.

E.1 Controlling State

The manuscript rewrite is based on the project state whose canonical phase is `VSG_LIMITED_SECTION_PROSE_REVIEW_PASSED_ARTIFACT_ONLY_NO_NEW_CLAIMS`. That state records a passed limited-section prose review, a passed claim-scope lint with zero violations, and zero enabled allowlist entries. The writing stage is artifact-only: it does not add generation, model scoring, Slurm submission, training, or new positive-result claims.

E.2 Evidence Artifacts

The main evidence groups used by this manuscript are:

- the trace-bound controllability corpus, which reports Qwen protected 94/96 and Llama protected 96/96 with all listed controls at 0/96;
- the public final-text predicate summary, which reports `codeword_recovered_blocks_total = 0`;
- the template leakage audit, which records the Qwen and Llama shallow AUC and duplicate/leakage quality gates;
- the public-predicate attack ladder, including rejection sampling, rewrite-lite, distillation-lite, and guided rewrite/graft;
- the ownership scenario stress test, which records 63 scenario-method cells, one supported cooperative trace-bound scenario, and zero supported public final-text-only portable scenarios.

E.3 Claim-Scope Contract

Every artifact should be interpreted under the following contract. Trace-bound codeword recovery is a provider-side event-access diagnostic. Public final-text predicates are observability and spoofing diagnostics. Source- mismatch accepted rows are not protected success. Current artifacts do not claim public text-only verification success, natural evidence success, phrase-decoder success, sanitizer robustness, payload-diversity evidence, model-family general verification, cryptographic provenance, or ownership proof.

E.4 Future Release Requirements

A public supplemental release should include anonymized artifact manifests, hashes for the relevant corpus summaries, the public predicate implementations, the attack scripts, and the claim-scope linter configuration. If future work turns the trace-bound route into a cooperative audit protocol, it should also release a trace-binding specification and verifier that checks output hashes, tokenizer hashes, event positions, controller configuration hashes, wrong-key replay, and wrong-payload replay. Those components are not claimed by this manuscript snapshot.

F Asset and License Inventory

This appendix records the non-model assets used by the current manuscript snapshot. It is an inventory for reproducibility and release review; it does not create and do not claim ownership proof or a public final-text verification claim.

The figures in this manuscript are generated from local CSV summaries rather than from third-party images. The manuscript therefore does not rely on stock art, screenshots, or externally licensed visual assets for Figures 1–5.

G Reproducibility Commands

This appendix records the commands used to rebuild the current paper assets from existing artifacts. The commands are local paper-asset commands only: they do not submit Slurm jobs, do not run generation, do not run model scoring, and do not train models.

Asset group	Source	Current use	Release note
Figure data CSVs	Project-generated VSG summary artifacts	Numeric inputs for Figures 3–5 and evidence tables	Release with hashes after anonymization review
Manuscript figures	Rendered locally from project CSVs by the VSG figure renderer	Static PNGs included in the manuscript	No external image assets used
Attack example excerpts	Existing source-mismatch attack artifacts	Report-only examples in Appendix C	Release only with source-mismatch labels and caveats
Bibliographic metadata	Public bibliographic records and cited papers	References and related-work positioning	Paper PDFs are not redistributed by this project
NeurIPS style file	Existing manuscript template file	Formatting only	Governed by the conference style package terms

Table 11: Asset inventory for the current manuscript snapshot. The table records provenance of paper assets and release notes; it is not a claim about model provenance.

G.1 Figure Rendering

```
python3 scripts/verification_substrate_gap/render_vsg_manuscript_figures.py
```

The renderer reads:

```
results/verification_substrate_gap/paper_figure_data_20260530/
```

and writes PNG figures to:

```
manuscripts/69db2644566dcc36c9da320e/figures/
results/verification_substrate_gap/paper_manuscript_figures_20260531/
```

G.2 Attack Example Review

```
python3 scripts/verification_substrate_gap/build_vsg_attack_examples_review.py
```

The script reads existing guided rewrite/graft examples and writes a compact review table under:

```
results/verification_substrate_gap/paper_attack_examples_20260531/
```

The output rows remain source-mismatch examples. They are report-only spoofing evidence and are not protected success or codeword recovery.

G.3 Claim-Scope Lint

```
MANUSCRIPT=manuscripts/69db2644566dcc36c9da320e
python3 scripts/verification_substrate_gap/lint_claim_scope.py \
  $MANUSCRIPT/main.tex \
  $MANUSCRIPT/section_01_introduction.tex \
  $MANUSCRIPT/section_02_related_work.tex \
  $MANUSCRIPT/section_03_problem_setup.tex \
  $MANUSCRIPT/section_04_tokenizer_alignment.tex \
  $MANUSCRIPT/section_05_bucket_level_injection.tex \
  $MANUSCRIPT/section_06_deterministic_verification.tex \
  $MANUSCRIPT/section_07_experiments.tex \
  $MANUSCRIPT/section_08_discussion_limitations.tex \
  $MANUSCRIPT/section_09_conclusion.tex \
  $MANUSCRIPT/appendix/proofs.tex \
  $MANUSCRIPT/appendix/formal_substrate_gap.tex \
  $MANUSCRIPT/appendix/attack_examples.tex \
  $MANUSCRIPT/appendix/extended_related_work.tex \
  $MANUSCRIPT/appendix/reproducibility.tex \
  $MANUSCRIPT/appendix/asset_licenses.tex \
  $MANUSCRIPT/appendix/reproducibility_commands.tex
```

G.4 PDF Build and Log Scan

```
cd manuscripts/69db2644566dcc36c9da320e
```

```
latexmk -pdf -interaction=nonstopmode main.tex
rg -n "undefined|Citation .* undefined|LaTeX Warning: Reference|\
LaTeX Warning: Citation|Fatal|Emergency stop|! LaTeX Error|\
There were undefined" main.log
```

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction state the verification-substrate-gap thesis, the trace-bound diagnostic scope, the public final-text codeword-recovery failure, and the source-mismatch spoofing interpretation. They explicitly withhold public text-only verification success and ownership-proof claims.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 8 discusses output transformations outside the guarantee, probe-free distillation, verifier spoofing boundaries, tokenizer dependence, and reported model-scale limitations.

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Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: Sections 3–4 state the substrate taxonomy and first-divergence diagnostic assumptions. Appendix A provides the first-divergence proof sketch; the paper does not claim a universal impossibility theorem.

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- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [No]

Justification: Appendix E reports tokenizer audit rules, protocol parameters, training settings, verifier settings, FAR calibration protocol, and requested compute budgets, but the current manuscript does not yet include complete end-to-end run commands and environment specifications for every reported artifact.

Guidelines:

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- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
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   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [No]

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   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
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6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [No]

   Justification: Appendix E specifies the VSG artifact scope and evidence groups, but exact public release manifests, hashes, commands, and environment specifications remain outside the manuscript.

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   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [No]

   Justification: Several scaling tables report Wilson intervals, but the manuscript does not yet provide uniformly defined uncertainty estimates for every baseline and negative-control comparison.

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- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
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8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

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   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

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